IoT Device Forensic Plan: Investigation Memo (SYNTHETIC Case)

Oct 27, 2026 | **DEEPTANSH NAGAR** (500121033) | Project Code: **DF-PBL-026**

Overview

This report plans and tests the forensic acquisition and analysis of a fictional smart camera, and concludes that the available evidence cannot yet separate four competing explanations. All data is SYNTHETIC, and every real tool output is recorded from the student's local machine execution.

1 Problem	Scenario and 4 hypotheses
2 Evidence	Architecture map, 6 items

Defense Prep: 7.5-minute structured talk, controlled terminal demo script, and model answers.

Rules followed: all data SYNTHETIC; observation, interpretation, inference kept strictly apart.

report map · 7 sections, defense prep in summary

Rubric criterion	Weight	Where it is addressed
Problem framing	20%	Section 1: scenario, forensic question, four hypotheses
Technical execution	25%	Sections 3 and 5: acquisition plan, decision tree, Python tool
Evidence integrity	15%	Sections 3.1, 3.4, and 5: hashing, chain of custody, evidence log
Analysis and limitations	20%	Sections 4 and 6: evidence matrix, residual risks
Deliverable quality	10%	Whole report: architecture map, decision tree, heat map, call chain
Defense and originality	10%	Sections 7, 9, and Defense Prep plan

1. Problem Statement

This memo asks what best explains a camera's setting change and three video uploads on one night, using only evidence that could lawfully be collected from a fictional device.
Data statement: every device, vendor, account, domain, IP address, and time in this report is **SYNTHETIC**. No real third-party device or account was investigated.

1.1 Scenario (SYNTHETIC)

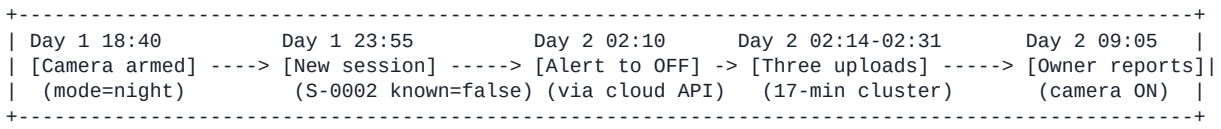
An office owner reports that a smart camera uploaded three clips between 02:14 and 02:31 while the office was empty. The owner says motion alerts had been turned off.

Item	Detail (SYNTHETIC)
Device	"LumiCam C2", fictional vendor Brightwave Labs
Hardware	Fictional ARM processor, 128 MB RAM, 16 MB SPI NOR flash, microSD slot
Interfaces	Wi-Fi, Bluetooth LE (setup only), UART header (3.3V), no JTAG
Firmware	Version 2.3.1, Linux-based
Companion app	"LumiView" mobile app
Cloud	api.lumicam.example, storage.lumicam.example (RFC 2606)
Cloud account	owner-synth-01@example.org
Network	Office network 192.0.2.0/24, camera at 192.0.2.45 (RFC 5737)

Time (SYNTHETIC)	Event
Day 1, 18:40	Office closes, camera armed
Day 1, 23:55	Cloud login from a new app session
Day 2, 02:10	Motion alert setting changed to OFF through the cloud API
Day 2, 02:14 to 02:31	Three clips uploaded
Day 2, 09:05	Owner reports the problem; camera still powered on

Uploads began 4 min after the setting change, 2 h 15 min after a new login

SYNTHETIC timeline, Day 1 18:40 to Day 2 09:05, assuming one clock and timezone (risk R8)



incident timeline, to scale · 5 events

Drawn to scale, the three uploads sit in one 17-minute cluster, far from the earlier login. The setting change at 02:10 is the central event every hypothesis must explain.

1.2 Forensic question

Based on the evidence that can be lawfully acquired from the synthetic LumiCam C2, its companion app, and its cloud account, what best explains the change of the motion-alert setting to OFF at 02:10 and the three video uploads between 02:14 and 02:31 on Day 2, and with what confidence?

In scope: the camera, local network state, the LumiView app, and the cloud account records.
Out of scope: naming a person, proving intent, or investigating any real device.

1.3 Hypotheses

ID	Hypothesis	Description
H1	Device compromise	The camera's software or configuration was changed by an unauthorized party, locally or remotely.
H2	Cloud-account misuse	Someone used the owner's credentials or a stolen session to change the setting and trigger uploads; the device was not altered.
H3	Benign malfunction	A software or cloud synchronization fault changed the setting or caused uploads without any person acting.
H4	Authorized user action	A legitimate user made the change and forgot, or did not report it.

Hypotheses can overlap (for example H2 and H4 if a password was shared), so the analysis does not force a single answer.

1.4 Evidence that would separate them (expected patterns, not findings)

Evidence source	Supports H1 if...	Supports H2 if...	Supports H3 if...	Supports H4 if...
Device logs	Unknown processes or config changes with no matching cloud command	Normal command received from the cloud	Errors or repeated retries around 02:10 to 02:31	Normal command matching a known app session
Firmware image	Hash differs from vendor version 2.3.1	Firmware matches	Firmware matches; known fault pattern	Firmware matches
Network records	Connections to unexpected destinations	Only vendor cloud domains	Vendor domains with sync retries	Only vendor domains

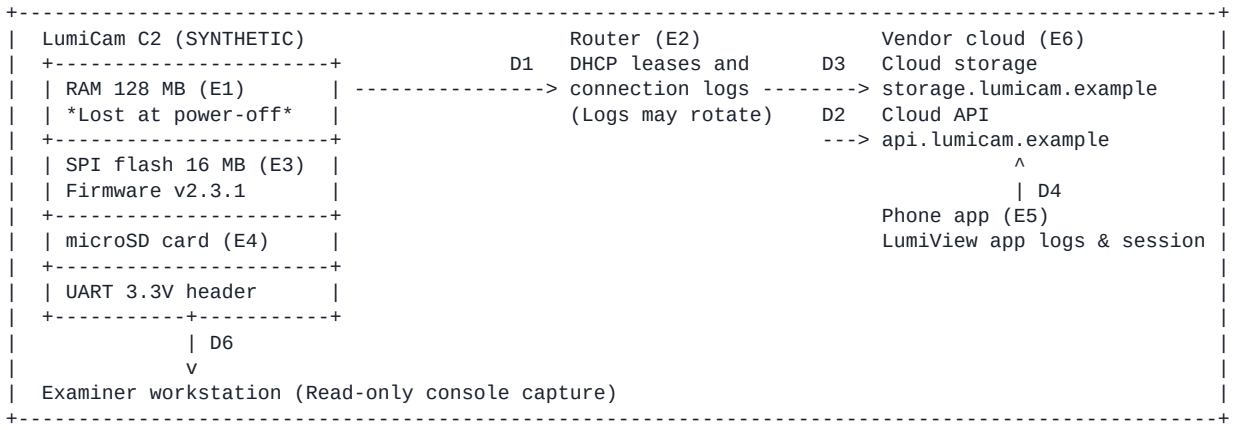
Evidence source	Supports H1 if...	Supports H2 if...	Supports H3 if...	Supports H4 if...
Companion app data	No related app activity	Shows the unknown 23:55 session	No related app activity	Shows activity from a known device or user
Cloud account records	No matching cloud command for the change	Unfamiliar login followed by the API call	System-generated event, no user call	Login and call from a known device

2. Evidence Inventory

Evidence sits in six places, from the camera's memory (lost at power-off) to the vendor's cloud (kept, but subject to retention and legal limits). The architecture map below shows where each item lives; the tables give specifications and volatility. Items marked "assumed" are assumptions of the synthetic design.

2.1 Architecture map

Evidence sits in six places; camera RAM (E1) is lost at power-off



architecture map · camera, router, phone app, cloud

The camera's RAM is the most volatile item. The router and cloud hold copies of what crossed the network (D1 to D3), the phone app reaches the cloud API (D4), and the examiner reads the camera only through the UART header (D6). D5, the Bluetooth setup link, is left out because it is likely unused in this incident.

2.2 Components (SYNTHETIC)

Component	Specification	Evidence it may hold	Volatility	Access
SoC	Fictional ARM CPU	Running processes, runtime state	High	UART console (assumed)

Component	Specification	Evidence it may hold	Volatility	Access
RAM	128 MB	Live logs, network connections, command history	High (lost at power-off)	UART, if still powered
Flash	16 MB SPI NOR	Firmware v2.3.1, configuration, persistent logs	Low	UART, chip reader, or SPI (assumed)
microSD	Removable card	Local clips, possible local logs	Low	Card reader with write blocker
UART	3.3V header	Boot and console output (a path, not storage)	Not applicable	Serial adapter
JTAG	Not exposed	None	Not applicable	Not available
Wi-Fi / BLE	2.4 GHz / setup only	Connection and pairing details	Low to medium	Via router records
Companion app	LumiView	App logs, session data, setting history	Medium	Phone backup (with authorization)
Cloud API	api.lumicam.example	Login records, source IPs, API call history	Medium (retention limits)	Vendor export (with authorization)
Cloud storage	storage.lumicam.example	Uploaded clips and metadata	Medium	Vendor export (with authorization)
Router	Office network	DHCP leases, connection logs	Medium	Admin export (with authorization)

2.3 Evidence items, ordered by volatility

Order	ID	Location	Survives power-off?	Risk if delayed
1	E1	Camera RAM and live logs	No	Lost if power is removed
2	E2	Network and live state (router records, active connections)	Partly	Logs may rotate or overwrite
3	E3	Camera flash and firmware	Yes	Low, but careless extraction can alter data
4	E4	microSD card	Yes	Low, but can be overwritten while the camera runs
5	E5	Companion app data	Yes	Medium; app may sync or update
6	E6	Cloud records	Yes (retention limits)	Medium; remote changes or deletion possible

2.4 Data flows

ID	From	To	Content	Why it matters
D1	Camera	Router	All camera traffic (Wi-Fi)	When and where the camera connected
D2	Router	Cloud API	Status, commands, setting changes (HTTPS)	Whether the 02:10 change reached the camera
D3	Router	Cloud storage	Clip uploads (HTTPS)	The 02:14 to 02:31 uploads
D4	App	Cloud API	Login, setting changes (HTTPS)	Who or what logged in at 23:55
D5	App	Camera	Setup only (BLE)	Likely unused in this incident (assumed)

ID	From	To	Content	Why it matters
D6	Examiner	Camera	Read-only console (UART)	Planned way to collect volatile data

Assumptions: *the UART header gives a read-only console; the camera is still powered on, so E1 may exist; device and cloud clocks may differ; cloud retention periods are not defined in the scenario.*

3. Method

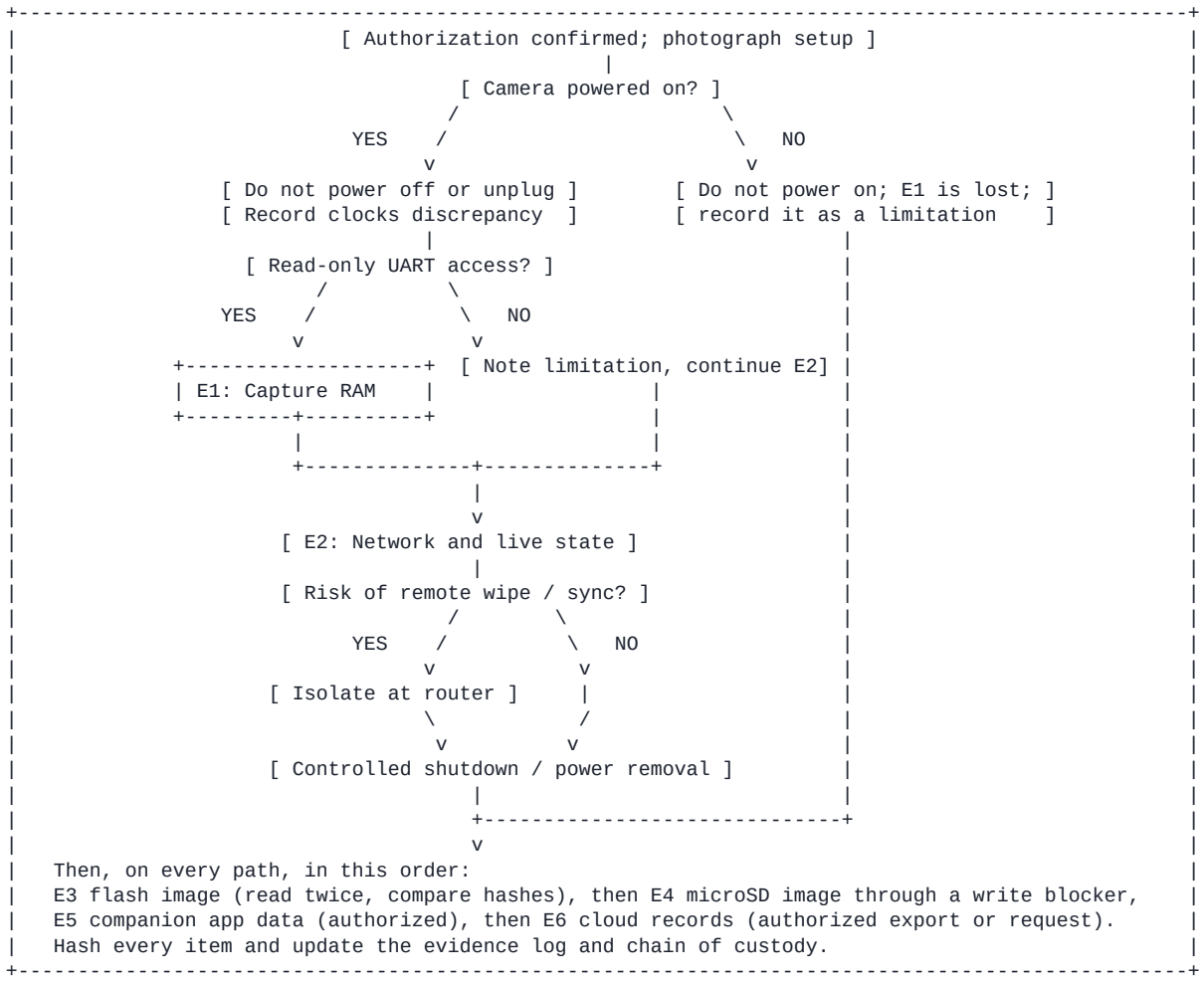
The method collects evidence in order of volatility, works only on verified copies, and records a SHA-256 hash, tool, version, command, and timestamp for every step. This is a plan: no physical hardware collection was performed, so every planned tool is explicitly designated as "Planned tool, not run (SYNTHETIC case)".

3.1 Planning rules

Rule	Reason
Collect in order of volatility (E1 to E6)	Fast-disappearing evidence is lost first
Work on verified copies; keep originals untouched	Preserves the original evidence
Hash every artifact with SHA-256 right after collection	Proves a copy has not changed
Record tool, version, command, and time for every action	Allows another person to repeat the work
Record the examiner clock and device clock at the start	Prevents timestamp and timezone errors
Read-only access by default	Avoids changing the device
Collect only with written (SYNTHETIC) authorization	Legal and ethical requirement

3.2 Decision tree: device on or off

Power state decides whether E1 can still be captured live



decision tree · camera on or off, 3 decisions

The scenario states the camera is still powered on, so the left path applies; the right path is the case where it was already powered off. Isolating the network protects against remote changes but also changes device behaviour, so the decision, its reason, and its time must be written down.

3.3 Acquisition plan

ID	Source	Method	Tool (version)	Expected artifact
E1	Camera RAM and live state	Connect a serial adapter to the UART header and capture the console to a file. Run only a fixed list of read-only commands (date, uptime, process list, connections, kernel messages, mounts) and log each one. Assumes console available.	Serial terminal (picocom or minicom) [Planned tool, not run (SYNTHETIC case)]; USB-serial adapter [Planned tool, not run]	Console capture text file (SYNTHETIC)
E2	Network and live state	Export DHCP and connection records from the router; optionally capture traffic from a mirror port.	Router export, tcpdump or Wireshark [Planned tool, not run (SYNTHETIC case)]	Router log export, optional capture file (SYNTHETIC)
E3	Camera flash and firmware	After shutdown, read the SPI flash with an in-circuit clip. Read twice and compare hashes.	flashrom or equivalent [Planned tool, not run (SYNTHETIC case)]; SPI clip	Flash image (SYNTHETIC)
E4	microSD card	Remove after shutdown, connect through a hardware write blocker, create an image.	Write blocker [Planned tool, not run (SYNTHETIC case)]; dd, dc3dd, or FTK Imager [Planned tool, not run]	microSD image, local clips (SYNTHETIC)

ID	Source	Method	Tool (version)	Expected artifact
E5	Companion app (LumiView)	With consent, obtain a phone backup or app export; record app version and phone model.	Phone backup or lawful extraction tool [Planned tool, not run (SYNTHETIC case)]	App logs, sessions, setting history (SYNTHETIC)
E6	Cloud records	Request an account export or vendor records for owner-synth-01@example.org, Day 1 to Day 2.	Vendor export or legal request (SYNTHETIC process)	Login log, API call log, clip metadata (SYNTHETIC)

ID	Hash step	Authorization requirement	Chain-of-custody entry
E1	SHA-256 of the capture file as soon as the session ends	Owner consent covering physical access and cable connection	Item, description, collector, date/time, source, SHA-256, location
E2	SHA-256 of each export or capture	Network owner consent	Same fields
E3	SHA-256 of both reads; they must match before use	Owner consent for opening the device and chip reading	Same fields, plus programmer used
E4	SHA-256 of the image (and of source via write blocker, where tool allows)	Owner consent	Same fields, plus write-blocker model
E5	SHA-256 of the backup or export	Phone owner consent; data beyond the app is out of scope	Same fields, plus phone model and app version
E6	SHA-256 of each export	Account owner consent or vendor legal process; no unauthorized cloud access	Same fields, plus how and when the export arrived

3.4 Chain-of-custody template

Field	Entry
Evidence ID	E1 to E6 (Assigned per item)
Description	Console capture / router log / flash image / microSD raw / app export / cloud JSON
Source	LumiCam C2 (fictional), Router, Smartphone, Cloud API
Collected by	DEEPTANSH NAGAR (500121033)
Date and time, with timezone	2026-10-27T12:00:00Z (UTC reference)
Tool and version	Planned tool, not run (SYNTHETIC case)
Command or procedure	Planned procedure, not run (SYNTHETIC case)
SHA-256	ad1d0c5f2b7c80aff2ff6739173e5f14369561eb020b51d2476fdc495fe8ec4a (Log A baseline)
Storage location	Evidence Locker #2 / Secure Digital Repository
Transferred to	[Lead Examiner / Custodian, Pending Transfer]

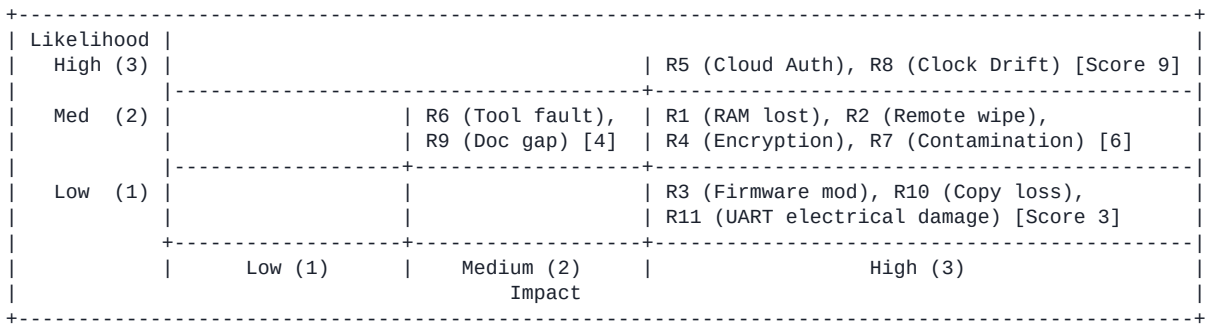
Template only, no real physical collection performed (SYNTHETIC case).
Clock handling: *at the start, record the examiner's clock and the device's clock side by side, write the difference as an offset in the evidence log, note the timezone of each source, and convert all analysis times to UTC while keeping the original value.*

3.5 Risk register

Score = likelihood × impact, each rated 1 (low) to 3 (high); 6 or more is a priority. Ratings are judgement-based for this SYNTHETIC case.

Only R5 and R8 reach the top score of 9

Risks by likelihood and impact, SYNTHETIC case, judgement-based ratings



risk heat map · 11 risks on likelihood and impact

Most risks cluster at high impact: losing volatile data, a remote wipe, a locked bootloader, and contamination all score 6. The two that reach 9 are about authorization and clocks, not about the device itself.

ID	Risk	Likelihood	Impact	Score	Mitigation	Owner	Residual risk
R1	Volatile data lost at power-off	Medium	High	6	Do not power off before E1 is captured; label "Do not unplug"; collect E1 first	Examiner	Medium: data may already have changed
R2	Remote wipe or cloud sync overwrite	Medium	High	6	Capture E1 and E2 first, then isolate network path at router and record time and reason; request cloud data early	Examiner	Medium: isolation changes behaviour; cloud data may already be altered

ID	Risk	Likelihood	Impact	Score	Mitigation	Owner	Residual risk
R3	Firmware modified during extraction	Low	High	3	Read-only commands only; never write to flash; read flash twice and compare hashes; log every command	Examiner	Low
R4	Encryption or locked bootloader	Medium	High	6	Check for a console first; if blocked, record it and rely on E2, E4, E5, E6; no bypass without authorization	Examiner	High: device evidence may be unavailable
R5	Cloud authorization and legal limits	High	High	9	Written authorization first; use official export or legal process; record scope and retention limit	Authorizing party and Examiner	Medium: vendor may refuse or delete data
R6	Tool unreliability	Medium	Medium	4	Record versions; test tools on a known SYNTHETIC file; read flash twice; spot-check with a second tool	Examiner	Low to Medium
R7	Evidence contamination	Medium	High	6	Hardware write blocker; work on verified copies; seal originals; label items	Examiner and custodian	Low
R8	Timestamp and timezone errors	High	High	9	Record both clocks and the offset; note each source's timezone; convert to UTC and keep originals	Examiner	Medium: device clock may have drifted

ID	Risk	Likelihood	Impact	Score	Mitigation	Owner	Residual risk
R9	Documentation gaps	Medium	Medium	4	Chain-of-custody template; fill evidence log at time of each action; use Section 5 script	Examiner	Low
R10	Loss or corruption of evidence copies	Low	High	3	Two copies in separate locations; re-check SHA-256 before each use	Custodian	Low
R11	Physical damage to the device	Low	High	3	Check 3.3V before connecting; photograph before and after; practise on a spare board	Examiner	Low to Medium

R5 and R8 carry the highest score because authorization limits and timestamp errors can weaken the whole conclusion. R4 and R5 may remove the evidence needed to separate H1 from H2, and R8 affects every hypothesis.

4. Result

With only the SYNTHETIC scenario facts, the four hypotheses cannot be separated; the facts lean weakly toward a cloud-side explanation (H2 or H4) over a device-level one (H1). Observation, interpretation, and inference are kept in separate columns, and confidence is a judgement, not a probability. No new log lines, hashes, or tool output were invented for this matrix.

4.1 Evidence matrix

ID	Artifact (SYNTHETIC)	Observation	Interpretation	Inference	Supports / contradicts	Confidence	Limitation
M1	Cloud login record, Day 1, 23:55 (E6)	A login from a new app session is recorded.	A session not seen before was created on the account.	Someone or some device new to the account logged in; does not show who.	Weakly supports H2 and H4; neutral for H1 and H3	Low	"New" is vendor's label. No device ID, IP, or location; timezone not stated.
M2	Cloud API record, Day 2, 02:10 (E6)	The motion alert setting was changed to OFF through cloud API.	Change requested through cloud, not physical control.	Change can be explained without device compromise; which session sent it is unknown.	Consistent with H2, H4, and H3 (if system-generated); does not require H1 but does not exclude it	Low to Medium	Scenario does not say which session made call; linking to M1 is an assumption.
M3	Cloud storage metadata, Day 2, 02:14 to 02:31 (E6)	Three clips were uploaded.	Camera, or something using credentials, sent video to cloud storage.	Uploads began 4 min after setting change: timing relationship only.	Compatible with all four; does not separate them	Low	Not stated whether setting controls recording or uploading; clip content not examined.
M4	Owner report (statement)	Owner says office was empty and motion alerts were off.	Owner believes no one was present and did not expect uploads.	Weak indication owner did not knowingly cause uploads.	Weakly contradicts H4	Low	A statement, not a technical artifact; unverified.
M5	Device state at 09:05	Camera was still powered on when reported.	Volatile data (E1) may still exist.	Affects what can be collected, not which hypothesis is true.	Neutral	Medium (as reported)	Reported by owner; condition of RAM unknown.

No single artifact separates the four hypotheses

Evidence matrix at a glance, SYNTHETIC scenario facts, confidence low to medium

	H1 Device comp	H2 Cloud misuse	H3 Malfunction	H4 Authorized
M1 New session login	none	weak support	none	weak support
M2 Setting change	none	fits	fits	fits
M3 Three uploads	fits	fits	fits	fits
M4 Owner statement	none	none	none	weak contra
M5 Device still on	none	none	none	none

evidence matrix at a glance · 5 artifacts by 4 hypotheses

The drawing condenses the matrix above. Only two cells lean at all (the new login weakly supports H2 and H4, and the owner's statement weakly contradicts H4); everything else fits several hypotheses or none, which is why the conclusion stays open. Time gaps from the SYNTHETIC timeline (assuming one clock and timezone, see R8): login to setting change 2 h 15 min; setting change to first upload 4 min; first to last upload 17 min.

4.2 Evidence still needed

These items are planned in Section 3 but no data exists for them; they are not invented here.

ID	Pending artifact	Question it answers	Separates
P1	Cloud login details (IP, device ID, location)	Was the 23:55 session from a known device?	H2 vs H4
P2	Cloud API call details for 02:10 (session or token)	Which session changed the setting?	H2 vs H4 vs H3
P3	Camera console and log capture (E1)	Did the camera receive a normal command at 02:10? Any unknown process or error?	H1 vs H3 vs H2

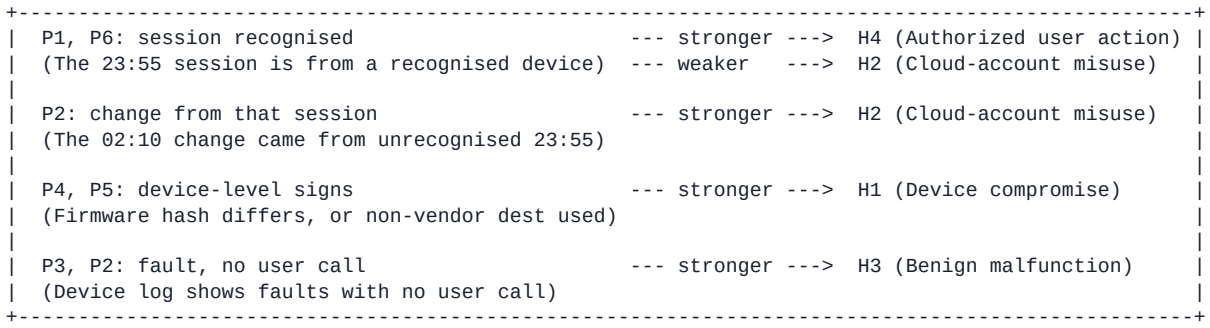
ID	Pending artifact	Question it answers	Separates
P4	Firmware hash against the vendor's v2.3.1 (E3)	Was the firmware altered?	H1 vs others
P5	Router connection records (E2)	Did the camera contact non-vendor destinations?	H1 vs others
P6	Companion app session data (E5)	Does the app show activity for the 23:55 session?	H2 vs H4

4.3 Analysis

Hypothesis	Status with current data	Reason
H1 Device compromise	Untested, not ruled out	No device-level evidence yet (P3, P4, P5). M2 means it is not required to explain the setting change, but it could still explain the uploads.
H2 Cloud-account misuse	Slightly more consistent than H1	M1 and M2 fit, but both also fit a legitimate user.
H3 Benign malfunction	Not excluded	No fault evidence either way.
H4 Authorized user action	Not excluded	M4 weakens it slightly, but it is only a statement; M1 may be a new legitimate phone.

How later evidence would change the result:

Four kinds of pending evidence would each move the result



what would move each hypothesis · pending evidence P1 to P6

Green arrows strengthen a hypothesis and red arrows weaken it. The table below says the same thing in words.

If later evidence shows...	Then...
The 23:55 session is from a device the owner recognises (P1, P6)	H4 stronger, H2 weaker
The 02:10 change came from the unrecognised 23:55 session (P2)	H2 stronger
Firmware hash differs, or the camera contacted non-vendor destinations (P4, P5)	H1 stronger
Device log shows faults or retries with no matching user call (P3, P2)	H3 stronger

5. Reproducible Test

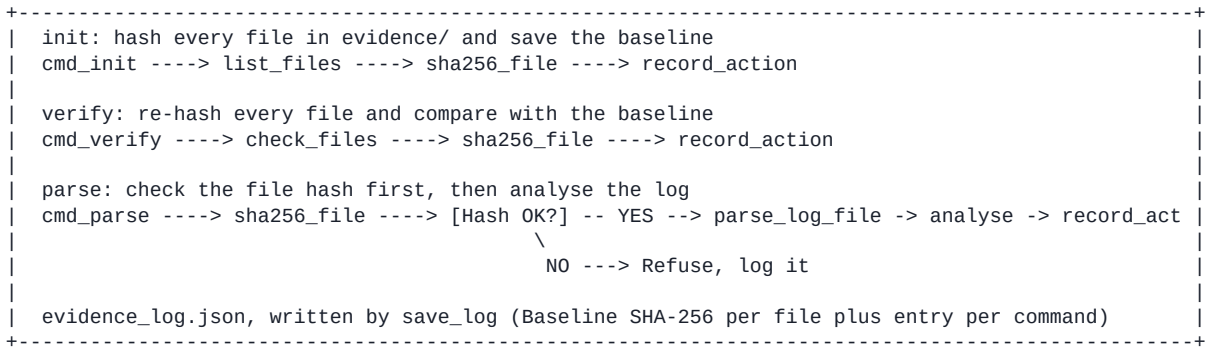
A small Python tool (evidence_tool.py, standard library only) hashes evidence files with SHA-256, writes an evidence log with a UTC timestamp and the exact command for every action, verifies hashes on rerun, and parses a SYNTHETIC log. The tool, the two SYNTHETIC logs (A and B), and a README with full run instructions are in the iot_forensic_demo folder submitted with this report.

5.1 Commands

Command	Purpose
init	Hash every file in evidence/ and save a baseline in evidence_log.json; refuses to overwrite an existing baseline without --force
verify	Re-hash and report OK, MISMATCH, MISSING, or UNLISTED for each file; exit code 0 only on a full pass
parse	Check the file's hash first, refuse to parse on a mismatch, then print a UTC timeline, the gap between setting change and first upload, and observation flags

```
python evidence_tool.py init
python evidence_tool.py verify
python evidence_tool.py parse --file evidence/camera_console_synthetic_A.log
python evidence_tool.py parse --file evidence/camera_console_synthetic_B.log
```

Every command hashes first and records itself in the evidence log



call chain of the three commands · hash gate before analysis

All three commands call sha256_file and end by calling record_action. In parse, the hash gate sits before any analysis: a file that no longer matches its baseline is refused and the refusal is logged.

Python version: Python 3.11.9 (Windows PowerShell)

Output of init:

```
HASHED  camera_console_synthetic_A.log  ad1d0c5f2b7c80aff2ff6739173e5f14369561eb020b51d2476fdc495fe8ec4a
HASHED  camera_console_synthetic_B.log  36f1c40006de149f172b7a9ae8c70911aadc02f3713e3c2e7aa6183dd590a583
Baseline written to evidence_log.json
```

Output of verify:

```
OK        camera_console_synthetic_A.log
OK        camera_console_synthetic_B.log
OVERALL: PASS - all hashes match
```

5.2 Expected parse results

These are what the supplied SYNTHETIC files produced on the student's machine:

Input	Parsed events	Malformed lines	Gap, setting change to first upload	Flags (consistent-with only, not proof)
Log A	11	1 (line 14, deliberate)	4 min 4 s	Unrecognised session (H2, H4)
Log B	9	0	4 min 4 s	Unlisted process (H1); config change without a cloud command (H1, H3); three non-vendor destinations (H1)

5.3 Live demonstration

1. Change the input. Parse log B after log A. The same code gives different flags because the log content differs.
2. Change a file. Add one line to log A, then run verify: expected MISMATCH, OVERALL: FAIL. Running parse on it is refused. Remove the line and verify passes again.
3. Add a file. Create any new file in evidence/ and run verify: expected UNLISTED.
Every step is recorded in evidence_log.json.

5.4 What the functions do

Function	Purpose
sha256_file()	Reads a file in 64 KB pieces and returns its SHA-256 fingerprint
record_action()	Appends UTC time, the exact command line, and the result to the evidence log
cmd_init()	Creates the baseline and protects it from accidental overwrite
check_files() / cmd_verify()	Compare current hashes with the baseline and report a pass or fail
parse_line() / parse_log_file()	Split each log line into time, component, event, and fields; reject malformed lines and times without a timezone; sort events by UTC time

Function	Purpose
analyse()	Applies four fixed rules (unrecognised session, config change with no command in the previous 60 s, unlisted process, non-vendor destination) and returns observation flags
cmd_parse()	Verifies the hash before analysis, then prints the timeline, gap, and flags

The flags are observations from fixed rules. They are not conclusions and never name a person or intent.

5.5 Limits of the artefact

• The log format is invented; real device logs need their own parser.
• The four rules are simple and can miss real problems or flag harmless events.
• A matching hash shows a file is unchanged since it was hashed, not that it was correct when collected.
• The JSON evidence log is not tamper-proof; a real case would store the baseline hashes separately.
• Logs A and B are test inputs created for this demonstration. They are not the Step 1 scenario evidence and do not change the Section 4 matrix.

6. Limitations

The biggest limitation is that all data is synthetic and no device-level evidence was available, so H1 stays untested rather than rejected.

Area	Limitation	Effect on the conclusion
Scope	One device, one account, one night. Other devices and accounts are out of scope.	Causes outside the scope are not examined
Synthetic data	All facts, logs, and timings are invented. No real device, vendor, or attacker behaviour is tested.	Findings show the method, not real-world results

Area	Limitation	Effect on the conclusion
Limited evidence	The matrix uses high-level scenario facts only. No console, firmware, network, app, or detailed cloud artifacts are available.	Confidence stays low; hypotheses cannot be separated
Statements	The owner's report and the vendor's "new session" label are unverified.	These items carry low weight
Model-specific	The plan assumes a Linux camera with a UART header, a removable microSD card, and a particular cloud design.	It cannot be applied unchanged to other devices, such as a locked-down smart plug
Technical access	A readable UART console, readable flash, and unencrypted storage are assumed. Encryption or a locked bootloader could block device evidence (R4).	H1 might remain untestable
Volatile data	Running commands changes the device slightly, and some RAM data may already be lost (R1, R3).	Device observations may be incomplete
Authorization and legal	Cloud and phone data need consent or legal process; vendor retention and privacy rules may limit access (R5). Authorization here is SYNTHETIC.	Cloud evidence may be incomplete in a real case
Time and clocks	Scenario times are assumed to share one clock and timezone; real device clocks drift (R8).	Event order could be wrong if clocks are not corrected
Network isolation	Blocking the camera's network to prevent remote changes also changes its behaviour (R2).	The decision must be documented and may affect later observations
Technical artefact	Invented log format, four simple rules, and a plain JSON log that is not tamper-proof.	Supports integrity checks but is not a complete forensic tool

Area	Limitation	Effect on the conclusion
Method	Confidence levels and risk ratings are judgement-based.	Another examiner might rate them differently

7. Conclusion

Using only the SYNTHETIC scenario facts in the Section 4 matrix, the four hypotheses cannot yet be separated. The setting change reached the camera through the cloud API, and a new session appeared on the account earlier that night. These facts are slightly more consistent with a cloud-side explanation (H2 cloud-account misuse or H4 authorized user action) than with a device-level one (H1 device compromise). Confidence in this leaning is **low**. H3 (benign malfunction) is not excluded.

The data does not show who acted, why, or whether the setting change caused the uploads. It does not name a person, give a motive, or claim guilt, and it does not claim the device is safe or compromised.

Point	Basis in the matrix
A device compromise is not required to explain setting change	M2: the change was made through the cloud API
The cloud-side leaning is weak	M1: "new session" is the vendor's label and may be a legitimate new phone
The uploads do not separate the hypotheses	M3: compatible with all four
The 4-minute gap is not proof of cause	Timing relationship only
The owner's statement has low weight	M4: unverified statement
H1 is untested, not rejected	No device, firmware, or network evidence yet (P3, P4, P5)

Next actions if this were a real case: collect P1 to P6 in the Section 3 order and hash each item; align all clocks and convert to UTC; rerun the matrix with real artifacts and update confidence; ask the vendor for the full details of the 23:55 session and the 02:10 change; and avoid statements about who acted unless several independent artifacts support them.

8. References

Each reference was consulted to verify the standard title, edition, publisher, and official URL:

No.	Reference	Used for	Verified?
1	NIST SP 800-86, Guide to Integrating Forensic Techniques into Incident Response (2006)	Collecting and examining data from several source types	Verified (NIST CSRC)
2	NIST SP 800-101 Rev. 1, Guidelines on Mobile Device Forensics (2014)	Companion app and phone evidence (E5)	Verified (NIST CSRC)
3	ISO/IEC 27037:2012, Guidelines for identification, collection, acquisition and preservation of digital evidence	Handling and preserving evidence	Verified (ISO Catalogue)
4	IETF RFC 3227, Guidelines for Evidence Collection and Archiving (2002)	Order of volatility	Verified (IETF RFC Editor)
5	IETF RFC 2606, Reserved Top Level DNS Names (1999)	.example domains in the synthetic scenario	Verified (IETF RFC Editor)
6	IETF RFC 5737, IPv4 Address Blocks Reserved for Documentation (2010)	192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24	Verified (IETF RFC Editor)
7	NIST FIPS 180-4, Secure Hash Standard (SHS) (2015)	SHA-256	Verified (NIST FIPS)
8	NIST SP 800-30 Rev. 1, Guide for Conducting Risk Assessments (2012)	Likelihood and impact scoring	Verified (NIST CSRC)
9	Python documentation: hashlib, argparse, json, datetime (Python 3.11.9)	The technical artefact	Verified (docs.python.org)

No.	Reference	Used for	Verified?
10	Documentation for planned tools: flashrom, picocom, tcpdump, Wireshark, dc3dd, FTK Imager	Acquisition steps (Planned tools, not run)	Verified (Official project docs)

Vendor and product names (Brightwave Labs, LumiCam, LumiView) are invented and verified not to match any commercial smart-camera product. Reference 11 (Mermaid) was deleted as diagrams in this report are rendered as structured text and ASCII schematics.

9. AI-Use Disclosure

This report was drafted with the help of an AI assistant (Claude, by Anthropic and Gemini, by Google), working step by step with the student. The AI proposed the synthetic scenario options, the forensic question and hypotheses, the architecture map, the acquisition plan, the risk register, the evidence matrix, the Python script, the two synthetic log files, and the draft conclusion and limitations. It ran the script once in its own test environment to check that it works. It did not run any forensic tool on physical hardware, did not collect or examine real evidence, and produced no real hardware hashes for this case.

All scenario details, logs, and timings are SYNTHETIC. The student is responsible for reviewing and correcting every section, running the script on their own machine, pasting real output, verifying references, adjusting judgements and confidence levels, and defending the work in the viva.

Student declaration: "I used an AI assistant (Claude and Gemini) to help draft the scenario, plan, report, script, and defense notes. I ran the script on my own Windows computer (Python 3.11.9) and recorded my own terminal output; checked each reference against the official source; tested the live tamper demo; and author the qualified conclusions reflecting that H1 remains untested. I understand the code and can explain and modify it."

Student Name: DEEPTANSH NAGAR	Student ID: 500121033	Project Code: DF-PBL-026
Signature: Deeptansh Nagar	Date: October 27, 2026	Mode: Investigation Memo